Scalability & Future Plan

Date	01 JULY 2026
Team ID	XXXXXX
Project Name	**HUMAN DEVELOPMENT INDEX(HDI)**
Maximum Marks	1 Mark

Scalability & Future Plan:

The HDI Predictor System has been designed with a modular architecture, making it scalable and adaptable for future enhancements. The current implementation successfully predicts the Human Development Index (HDI) using a Machine Learning model integrated with a Flask backend and a React frontend. Future improvements will focus on increasing system scalability, supporting larger datasets, improving security, and enhancing the user experience

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Uses a static HDI dataset without real-time updates	Predictions are limited to the available dataset and may not reflect the latest development indicators.	**High**
2	Prediction history is stored locally in the browser	Data may be lost if browser storage is cleared and cannot be shared across devices.	**Medium**
3	No user authentication or role-based access control	The application cannot support personalized user accounts or secure multi-user access.	**Medium**

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports local usage by a limited number of user	Deploy the backend on **AWS EC2** with **Gunicorn** and **Nginx** to handle multiple concurrent users
2	Data Storage	Uses local CSV files and browser Local Storag	Integrate a relational database such as **MySQL** or **PostgreSQL** for centralized and persistent data storage

3	Performance	Single Machine Learning model with local execution	Optimize inference, implement caching, and use model serving techniques for faster response times
4	Security	Basic input validation	Add user authentication (JWT/OAuth), HTTPS, API security, and role-based access control

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Deploy backend on AWS EC2 and frontend on Vercel	Short Term	Public accessibility and cloud-based deployment
Phase 2	Integrate a MySQL/PostgreSQL database and user authentication	Medium Term	Secure data management and personalized user experience
Phase 3	Connect to real-time HDI data sources (UNDP/World Bank APIs) and add country comparison dashboards	Long Term	More accurate predictions and advanced analytical capabilities
Phase 4	Develop a mobile application, multilingual support, AI chatbot assistance, and interactive geographical visualizations	Future Release	Improved accessibility, user engagement, and scalability